

Yi Hu

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EDUCATION

PhD in Biomedical Engineer

University of Alberta

Jan. 2021 – Current

Edmonton, Canada

MSc in Control Theory and Control Engineering

Sichuan University

Sept. 2017 – Jun. 2020

Chengdu, China

B.S. in Process Equipment and Control Engineering

Hefei University of Technology

Sept. 2013 – Jun. 2017

Hefei, China

RESEARCH EXPERIENCE

Human-Centered Autonomy Lab

Graduate Research Assistant, University of Alberta

Jan. 2024 – Present

Edmonton, Canada

- Enhanced large Vision-Language-Action (VLA) models by integrating a 3D spatial representation module, enabling cross-modal fusion for geometrically grounded tasks. Designed a Mixture-of-Experts action head to specialize in spatial reasoning. Fine-tuned models (Pi0, OpenVLA) on real-world robotic datasets and deployed the optimized system for real-time RGB-D inference and task execution on a Franka manipulator.
- Architected a single-step diffusion framework to generate temporally structured motion primitives, enabling reactive, high-speed manipulation. Achieved a 7-10x inference speedup in simulation (MetaWorld, ManiSkills) and validated performance on physical hardware by successfully executing a dynamic ball-catching task with a Franka robot.
- Engineered a reinforcement learning framework to transfer motion priors from human activity demonstrations to robotic surgical tasks. Validated the framework on a da Vinci Surgical System, enabling autonomous, precise tool manipulation and optimized endoscopic camera control derived from human demonstration data.

Telerobotic and Biorobotic Systems Group

Graduate Research Assistant, University of Alberta

Jan. 2021 – Dec. 2023

Edmonton, Canada

- Developed a complete autonomous ultrasound scanning system for a Franka robot, integrating admittance control for compliant tissue contact and a force regulation policy. Trained and deployed a U-Net model for real-time ultrasound image assessment, creating a closed-loop framework for scan quality and probe positioning.
- Implemented an autonomous endoscopic camera control system for the da Vinci robot. Applied unsupervised learning to derive motion primitives from surgical demonstrations, generating optimized, occlusion-free camera trajectories within the constrained anatomical workspace.
- Engineered a collision-aware 6-DoF motion planning stack for the da Vinci Patient Side Manipulator. Developed algorithms for smooth trajectory generation with orientation constraints, enabling precise and safe tool manipulation in confined surgical environments.
- Built a robust teleoperation interface using 3D hand pose estimation via MediaPipe. Mapped 6-DoF human hand motions to robot end-effector commands with dynamic safety constraints and implemented gesture recognition for triggering discrete robotic actions during ultrasound probe manipulation.

Perception, Motion Control and Intelligent Robot Innovation Lab

Graduate Research Assistant, Sichuan University

Sept. 2017 – Dec. 2020

Chengdu, China

- Built an end-to-end event-camera SLAM system featuring event-to-frame conversion, Hessian-based tracking initialization, and keyframe-aware backend optimization for long-duration robotic inspection tasks.
- Designed robust nonlinear control algorithms for fixed-joint robotic manipulators, explicitly accounting for actuator saturation bounds and modeled external disturbances.
- Derived and implemented kinematic and dynamic models for a differential-drive mobile robot platform. Developed and tuned model-predictive trajectory tracking controllers to achieve precise path following and positioning.
- Developed a Unity-based simulation environment for vehicle wheel robot navigation and visual localization. The environment featured synthetic sensor feeds for camera-based visual localization algorithms.

SKILLS

Technical Skills: Python, MATLAB/Simulink, Pytorch, TensorFlow, ROS2, SolidWorks, AutoCAD, Ansys, Comsol
Languages: Mandarin(Native), Cantonese(Native), English(Fluent)